Appendix A

A Brief Matrix Primer

This section provides a brief introduction to matrices and vectors. You can learn more by consulting any introductory book on the subject (just about anything with a title along the lines of 'An Introduction to Matrices' will do), or by searching around on the Internet. Wikipedia http://en.wikipedia.org/wiki/Matrix_(mathematics) is not a bad place to start.

A.1 What is a Matrix?

A matrix is a collection of numbers (or 'scalars') arranged in a rectangular array. If this array has r rows and c columns then the matrix is called an r by c matrix. It has the form

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1j} & \cdots & x_{1c} \\ x_{21} & x_{22} & \cdots & x_{2j} & \cdots & x_{2c} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{i1} & x_{i2} & \cdots & x_{ij} & \cdots & x_{ic} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{r1} & x_{i2} & \cdots & x_{ij} & \cdots & x_{rc} \end{bmatrix} \tag{A.1}$$

so that x_{ij} denotes the i, jth *element* of the matrix X (i.e. the entry in the ith row, jth column). Sometimes you may see matrices denoted by the shorthand element notation $X = (x_{ij})$ or $X = [x_{ij}]$. Note that the matrix itself is usually denoted by an upper case letter while its elements are lower case (with subscripts).

Example A.1: A Simple Matrix

Consider the matrix A defined by

$$A = \begin{bmatrix} 3 & 4 \\ 2 & 2 \\ 4 & 1 \end{bmatrix}.$$

This is a 3 by 2 matrix. Its first row is $(3, 4)$ and its second column is $(4, 2, 1)$. The $1, 2$ element, a_{12}, is 4.

A.2 Some Special Types of Matrix

A.2.1 Square Matrices

If a matrix has the same number of rows as columns then it is called a square matrix. Square matrices have some nice properties that are not shared by non-square matrices, like the possession of an inverse (as we shall discuss below).

Example A.2: A Simple Square Matrix

The matrix A defined by

$$A = \begin{bmatrix} 1.74 & 4.11 \\ 3.11 & 3.16 \end{bmatrix}$$

is a square matrix. The *main diagonal* of this matrix contains the elements 1.74 and 3.16.

A.2.2 Symmetric Matrices

A matrix $X = (x_{ij})$ is symmetric if $x_{ij} = x_{ji}$ for all i and j. In other words, the portion of the matrix below the main diagonal is the mirror image of the portion of the matrix above the main diagonal if you hold a mirror along the main diagonal.

Example A.3: A Simple Symmetric Matrix

The matrix A defined by

$$A = \begin{bmatrix} 3 & -2 & 0 \\ -2 & 1 & 4 \\ 0 & 4 & 5 \end{bmatrix}$$

is a symmetric matrix.

A.2.3 Diagonal Matrices

A diagonal matrix has zero entries everywhere away from the main diagonal.

Example A.4: A Simple Diagonal Matrix

The matrix A defined by

$$A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

is a diagonal matrix.

A diagonal matrix is, of course, symmetric.

A.2.4 Identity Matrices

The identity matrix of size (or order) n is the $n \times n$ matrix with ones along the main diagonal and zero elements everywhere else. The identity matrix of order n is often denoted I_n (or simply I when the order is obvious and/or does not need to be specified).

Example A.5: Some Identity Matrices

The matrix

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

is the identity matrix of order 3.

Identity matrices can be thought of as the 'units' in matrix algebra, in that they play much the same role as the number 1 in standard arithmetic.

A.2.5 Vectors

A matrix with a single column is a *column* vector. A matrix with a single row is a *row vector*. Vectors are typically assumed to be column vectors unless explicitly stated otherwise.

Vectors are a special type of matrix, and have their own notation. Specifically, vectors are usually denoted by a bold, lower case character, with elements specified by a single subscript.

Example A.6: Some Vectors

The vector

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1.4 \\ 0.5 \\ -0.3 \end{bmatrix}$$

is a column vector.

The vector $\boldsymbol{v} = (4.2, 5.0)$ is a row vector.

A.3 Matrix Operations

A.3.1 Matrix Addition and Subtraction

Only matrices of the same size (i.e. with the same number of rows and columns) can be added or subtracted. When the matrices are of the same size, addition and subtraction are done element by element in the natural way. Hence if $A = (a_{ij})$ and $B = (b_{ij})$, then $C = A + B$ has i, jth element given by $c_{ij} = a_{ij} + b_{ij}$.

Example A.7: Some Simple Matrix Arithmetic

An example of addition:

$$\begin{bmatrix} 3 & 4 \\ 2 & 2 \\ 4 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 6 \\ 4 & 2 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 10 \\ 6 & 4 \\ 5 & 3 \end{bmatrix}.$$

An example of subtraction:

$$\begin{bmatrix} 3 & 4 \\ 2 & 2 \\ 4 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 6 \\ 4 & 2 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ -2 & 0 \\ 3 & -1 \end{bmatrix}.$$

A.3.2 Multiplication by a Scalar

Multiplication of a matrix by a scalar (i.e. a single number) is achieved by multiplying each element of the matrix by that scalar.

Example A.8: More Matrix Arithmetic

If

$$A = \begin{bmatrix} 3 & 4 \\ 2 & 2 \\ 4 & 1 \end{bmatrix}$$

then

$$4A = \begin{bmatrix} 12 & 16 \\ 8 & 8 \\ 16 & 4 \end{bmatrix}.$$

A.3.3 Matrix by Matrix Multiplication

For matrices A and B we can evaluate the produce AB only if the number of columns of A is the same as the number of rows of B. If the matrices are conformable in this way, then the i, jth element of the product $C = AB$ is given by

$$c_{ij} = \sum_k a_{ik} b_{kj}.$$

In other words, to evaluate the i, jth element of the product we first take each element on the ith row of A and multiply it by the corresponding element on the jth column of B, and then sum up all these individual products.

Example A.9: A Simple Matrix Product

Define

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 3 & 4 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 3 & 4 \\ 2 & 2 \\ 4 & 1 \end{bmatrix}$$

Since the number of columns of A is equal to the number of rows of B (three in each case) we can multiply these matrices to give $C = AB$.

Now, the element in position $1, 1$ in C is obtained by combining the first row of A with the first column of B as described above. Specifically, we have

$$c_{11} = \sum_{k=1}^{3} a_{1k} b_{k1} = 2 \times 3 + 1 \times 2 + 3 \times 4 = 20.$$

Moving on to c_{12}, this is obtained by working with the first row of A and the second column of B to give

$$c_{12} = 2 \times 4 + 1 \times 2 + 3 \times 1 = 13.$$

The full result of the matrix multiplication is

$$AB = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 3 & 4 \end{bmatrix} \begin{bmatrix} 3 & 4 \\ 2 & 2 \\ 4 & 1 \end{bmatrix} = \begin{bmatrix} 20 & 13 \\ 25 & 14 \end{bmatrix}.$$

As an exercise you should check that you can derive the entries on the bottom row.

One important aspect of matrix multiplication is that it is non-commutative. This means that the order matters in the multiplication. For standard numbers multiplication is commutative; for example $2 \times 3 = 3 \times 2$, it does not matter which way around we arrange the 2 and the 3. However, for matrices the product AB will typically be different to BA.

Example A.10: A Simple Matrix Product Revisited

Consider the matrices from the previous example. We will now compute the product BA. We have

$$BA = \begin{bmatrix} 3 & 4 \\ 2 & 2 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 3 \\ 1 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 10 & 15 & 25 \\ 6 & 8 & 14 \\ 9 & 7 & 16 \end{bmatrix}.$$

This is a different result to that in the previous example. Indeed, the matrix BA is even a different size to AB in this case (3×3 versus 2×2).

You should check that you can derive the matrix product in this example. As a starter, note that the top left element of the product is given by $3 \times 2 + 4 \times 1 = 10$.

To recap, you cannot generally change the order of matrices when forming products[1].

Matrix by matrix multiplication is perhaps the aspect of matrix algebra that people find the most off-putting at first. Not only does it take a bit of getting used to, but also it is not immediately obvious why matrices should be multiplied in the way described above. Rest assured that there are some very good mathematical reasons for this method of matrix multiplication[2].

A.3.4 Product of Matrix by Vector

Multiplication of a matrix by a vector follows exactly the same rules as for matrix by matrix. Just recall that the vector can be thought of as a matrix with one row or column. However, if we use the single subscript notation to identify the elements of the vector then the equations look a bit different. Specifically, let $A = (a_{ij})$ be an $r \times c$ matrix, and let $\boldsymbol{v}$ be a column vector with c elements (i.e. a

[1] Note that addition and subtraction of matrices is commutative – you can change order for those operations.

[2] There are a number of possible methods for obtaining the product of two matrices, but the one that we describe is the most common.

$c \times 1$ matrix). Then if we denote the product by $A\boldsymbol{v} = \boldsymbol{x}$, we find that $\boldsymbol{x}$ is also a column vector but with r elements, defined by

$$x_i = \sum_{j=1}^{c} a_{ij} v_j.$$

Example A.11: A Simple Product

Let

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 3 & 4 \end{bmatrix}$$

and

$$\boldsymbol{v} = \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$$

Then

$$A\boldsymbol{v} = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 3 & 4 \end{bmatrix} \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 14 \\ 12 \end{bmatrix} = \boldsymbol{x}.$$

Note that the upper entry in the final vector was calculated as follows:

$$x_1 = 2 \times 4 + 1 \times 0 + 3 \times 2 = 14.$$

You should check that you can derive the lower entry, x_2.

A.3.5 Multiplication by the Identity Matrix

We introduced the identity matrix above, and noted that it plays the role of a 'unit' in matrix algebra. Now that we know how to perform matrix multiplication we can be a little more precise about this. Specifically, if the matrices and vectors involved are conformable (i.e. of the right sizes to permit the multiplications to be performed), then

$$\begin{aligned} AI &= A \\ IB &= B \\ I\boldsymbol{v} &= \boldsymbol{v} \end{aligned}$$

where A and B are matrices and $\boldsymbol{v}$ a vector.

A.3.6 Matrix Transpose

The transpose of a matrix is obtained by interchanging rows and columns. The transpose of A is denoted A^{T}.

Example A.12: Some Simple Transpositions

If A is defined by

$$A = \begin{bmatrix} 3 & 4 \\ 2 & 2 \\ 4 & 1 \end{bmatrix}$$

then its transpose is given by

$$A^{\mathsf{T}} = \begin{bmatrix} 3 & 2 & 4 \\ 4 & 2 & 1 \end{bmatrix}.$$

The row vector $\boldsymbol{v} = (4.2, 5.0)$ has transpose

$$\boldsymbol{v}^{\mathsf{T}} = \begin{bmatrix} 4.2 \\ 5.0 \end{bmatrix}$$

which is a column vector.

A useful property of transposition is as follows: $(AB)^{\mathsf{T}} = B^{\mathsf{T}}A^{T}$. In other words, the transpose of a product is the product of transposes in the reverse order.

Vector transposition proves useful in computing sums of squares in statistics. Specifically, for a column vector $\boldsymbol{v}$,

$$\boldsymbol{v}^{T}\boldsymbol{v} = (v_1, v_2, \ldots, v_n) \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = v_1 \times v_1 + v_2 \times v_2 + \cdots v_n \times v_n = \sum_{i=1}^{n} v_i^2.$$

A.3.7 Matrix Inverse

For a given matrix A, the inverse matrix is denoted A^{-1} and satisfies $AA^{-1} = I = A^{-1}A$. Only square matrices can have inverses, and even some square matrices will be 'uninvertible'[3]. Inversion of a matrix is analogous to finding the reciprocal when dealing with standard numbers.

Example A.13: A Simple Matrix and Its Inverse

[3]There are, in fact, a variety of 'generalized inverses' that can be applied even when the usual inverse does not exist

Consider the matrix A defined by

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix}.$$

The inverse of this matrix is

$$A^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{4} & \frac{3}{4} \end{bmatrix}.$$

As an exercise you should perform the matrix multiplication to confirm that

$$AA^{-1} = \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{4} & \frac{3}{4} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

and

$$A^{-1}A = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{4} & \frac{3}{4} \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I.$$

In general evaluation of a matrix inverse is a tedious matter that is best left to a computer[4]. However, there are certain situations in which matrix inverses can be computed easily by hand. For example, the inverse of a diagonal matrix is just another diagonal matrix with reciprocal elements on the main diagonal.

Example A.14: Inverse of a Diagonal Matrix

If A is defined by

$$A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

then

$$A^{-1} = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{5} \end{bmatrix}.$$

Another straightforward case is inversion of a 2×2 matrix. We have

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

As an exercise confirm that this formula produces the inverse A^{-1} in the previous Example.

[4] In my experience computers don't get bored easily...

A.4 The End of the Line

This is the end of what has been an express introduction to matrices and vectors. Don't worry too much if you are still feeling a bit at sea with matrices. We have covered a little more here than you need to know to pass 161.304. In terms of what is absolutely essential, you must understand enough about matrices and vectors to be comfortable with matrix calculations on Markov chain transitions and stationary distributions in Chapter 4. If you don't feel that you are at that stage then you should re-read this appendix, and/or consult a suitable book or on-line resource.